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ORIGINAL ARTICLE

Changes in muscle fascicles of tibialis anterior during anisometric contractions are not associated with motor-output variability of the ankle dorsiflexors in young and old adults

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Abstract This study examined the associations between the fluctuations of foot acceleration during shortening and lengthening contractions with the electromyographic (EMG) activity of lower leg muscles and ultrasound measures of tibialis anterior fascicle length and pennation angle. Young (24.9 ± 4.17 years) and old (74.8 ± 3.31 years) adults lifted and lowered a submaximal load with the foot at different speeds (3°/s–50°/s). The standard deviation (SD) of foot acceleration normalized to the load lifted was similar for young (12.2 ± 7.22 cm s^{-2}/kg) and old adults (14.3 ± 8.03 cm s^{-2}/kg; $P = 0.093$). The changes in tibialis anterior muscle fascicle length and pennation angle were similar for young and old adults ($P \geq 0.233$), but greater for shortening (fascicle length: 0.937 ± 0.633 cm, pennation angle: 1.61 ± 0.918°) than for lengthening contractions (fascicle length: 0.806 ± 0.521 cm, pennation angle: 0.966 ± 0.632°; $P \leq 0.014$). The changes in fascicle length and pennation angle were not associated with the SD of foot acceleration ($r^2 \leq 0.031$; $P \geq 0.092$). The surface EMG of tibialis anterior was greater for the shortening contractions than for the lengthening contractions ($P < 0.001$), but triceps surae EMG was similar for the two types of contractions ($P = 0.304$). The results suggested that the influence of movement speed on variability in performance was similar for shortening and lengthening contractions with the dorsiflexor muscles; furthermore, old adults were able to match the performance of young adults.

Keywords Muscle fascicles · Motor-output variability · Tibialis anterior · Shortening contractions · Lengthening contractions

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Introduction

The capacity of an older adult to achieve an intended muscle action is often less than that of a young adult, whether the task involves forceful contractions or contractions involved in a test of manual dexterity (Bazzucchi et al. 2004; Klass et al. 2008; Marmon et al. 2008; Yamauchi et al. 2009). The adaptations responsible for the declines in performance often exhibited by older adults include the loss of muscle mass associated with sarcopenia (Dey et al. 2009; Forrest et al. 2007; Goodpaster et al. 2006; Narici and Maffiuli 2010) and an increase in motor-output variability (Christou et al. 2007; Ofori et al. 2010; Sosnoff and Voudrie 2009; Vaillancourt et al. 2004). Although the consequences of sarcopenia and a worsening of motor-output variability are typically observed concurrently (Barry et al. 2007; Carville et al. 2007; Christou and Carlton 2002c; Christou et al. 2003; Graves et al. 2000; Seynnes et al. 2005; Tracy and Enoka 2002; Tracy et al. 2007), it is possible for one adaptation to appear without the other. For example, there can be no decline in the strength of a hand muscle, yet the same muscle can have a reduced ability to perform steady contractions (Burnett et al. 2000; Laidlaw et al. 2002; Poston et al. 2008a, b). However, it appears highly improbable for older adults to

experience a decrease in muscle strength without an increase in motor-output variability.

The influence of sarcopenia-related changes on motor-output variability has received relatively little attention. The typical approach has been to estimate sarcopenia with a surrogate measure, such as muscle strength or power, and to assess its influence on the ability of an individual to perform steady or accurate contractions. Some investigators, however, have characterized sarcopenia in terms of the length and orientation of muscle fascicles. For example, age-associated reductions in muscle fascicle length reduce the maximal shortening velocity of muscle (Narici et al. 2003; Narici and Maganaris 2006; Thom et al. 2007); a reduction in fascicle pennation angle is presumed to indicate a decrease in the force-generating capacity of muscle due to sarcopenia (Lexell et al. 1988; Narici et al. 2003; Narici and Maganaris 2006). Furthermore, changes in muscle fascicle length and pennation angle appear to vary with contraction speed and type, at least in young adults (Reeves and Narici 2003).

The purpose of this study was to compare a measure of motor-output variability with changes in tibialis anterior fascicle length and pennation angle as young and old adults performed submaximal shortening and lengthening contractions at different target speeds. Motor-output variability was quantified as standard deviation (SD) of foot acceleration. The hypothesis was that the SD of foot acceleration would be greater during lengthening contractions, and this would be accompanied by differences in muscle activity and changes in muscle fascicle length and pennation angle. Furthermore, it was expected that there would be a greater SD of foot acceleration for faster contraction speeds and for old adults, which would be accompanied by relatively greater changes in muscle fascicle length and pennation angle. Some of these data are presented in abstract form (Rudroff et al. 2009).

Methods

Shortening and lengthening contractions were performed by 13 young (24.9 ± 4.17 years; 7 men) and 14 old adults (74.8 ± 3.31 years; 8 women) with the dorsiflexor muscles of the left leg at target speeds of 3, 15, 30, and 50°/s. The acceleration of the foot and the EMG activity of tibialis anterior and triceps surae muscles were measured during the contractions. All subjects were right-footed as determined by the Waterloo Footedness Survey (Elias et al. 1998) and screened by a physician to ensure the absence of neurological disorders; they provided written informed consent before participating in the experimental protocol, which was approved by the Human Research Committee of the University of Colorado, Boulder.

Experimental set-up

The subjects sat on a chair with the upper leg horizontal and the lower leg vertical. Foam pads supported the left knee to prevent compression of the tibial and common peroneal nerves during the experimental protocol. A modified foot orthosis (Shift™, MSR) was attached to the left foot with the metatarsophalangeal joint of the big toe placed 14 cm from the distal end of the orthosis where the load was applied (Fig. 1a). A load of 15% of the one-repetition maximal (1-RM) load was hung from the front of the shoe, and was located at 27.9 ± 1.27 cm from the ankle. The angle between the foot and lower leg was measured with a flexible axis goniometer (Noraxon U.S.A. Inc., Scottsdale, AZ, USA; nominal output range ± 180°, accuracy ± 2°), which was attached across the ankle joint parallel to the lateral aspect of the lower leg and the foot. The acceleration of the foot in the direction perpendicular to the plane of the foot orthosis was measured with an accelerometer (Noraxon U.S.A. Inc., Scottsdale, AZ, USA; linear range: 981 m/s^2) attached to the orthosis over the metatarsophalangeal joint of the big toe.

The EMG signals for the tibialis anterior, medial and lateral gastrocnemius, and soleus were measured with Cleartrace REF 1700-001 silver-sliver chloride surface electrodes (ConMed® Corporation, Utica, NY, USA) attached to the surface of the skin in a bipolar configuration over the muscles. EMG signals had a differential gain of 500 ± 2%, a common mode rejection ratio >100 dB, and were band-pass filtered between 10 and 500 Hz. The surface electrodes for the tibialis anterior were attached to the skin over the proximal aspect of the muscle (Fig. 1a). A common ground electrode was placed over a bony prominence of the knee. The acceleration of the foot, ankle angle, and EMG activity of the muscles were acquired with a Telemyo2400 wireless system (Noraxon U.S.A. Inc., Scottsdale, AZ, USA) digitized at 1,500 Hz and stored on a computer.

Images of the tibialis anterior along the mid-sagittal plane were acquired using B-mode ultrasonography (Toshiba Powervision 5000) with an 8 MHz linear-array probe (Toshiba PLM-805AT). The ultrasound images were acquired using the image acquisition toolbox of MATLAB® R2008b (The MathWorks, Inc., Natick, MA, USA) at 30 frames/s. Images were obtained by placing the probe directly over the thickest portion of the muscle, and the probe's position was adjusted for each subject during the experiment to maintain a clear image of the muscle. A water-soluble gel was applied to the probe to allow for acoustic contact between the probe and the surface of the skin. Limited pressure was applied to the probe to ensure that the probe did not significantly depress the derma over the muscle. The probe was held in the same position on the skin over the muscle with an adjustable clamp (Flexbar, Islandia, NY, USA) and by one of the investigators during the contractions.

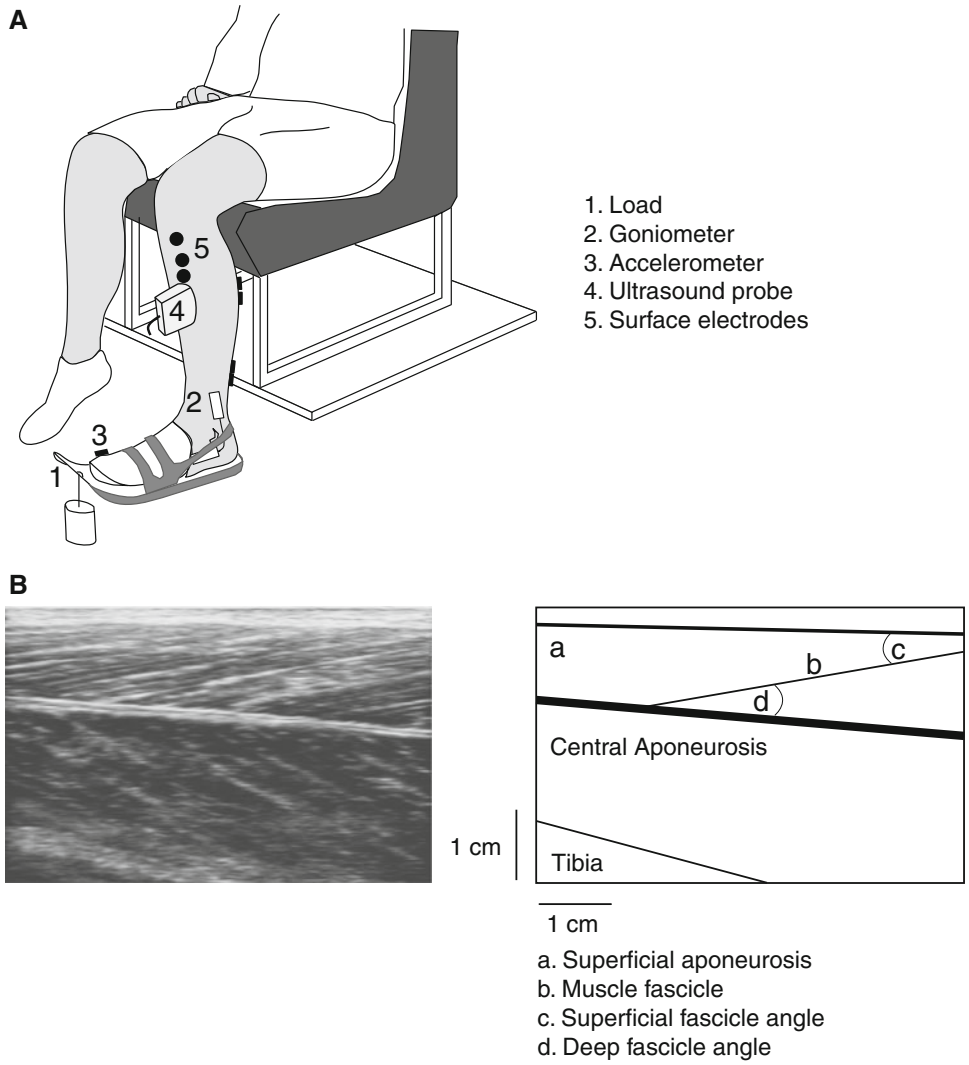

Fig. 1 **a** Schematic of the experimental set-up. The subject was seated in a chair with the *upper leg horizontal* and the *lower leg vertical*. *1* A load of 15% 1-RM was hung from the orthosis attached to the foot. *2* A goniometer was placed across the ankle joint to measure the angle between the foot and the lower leg. *3* Foot acceleration was measured in the direction perpendicular to the plane of the foot orthosis with an accelerometer attached to the orthosis at the level of the metatarsophalangeal joint of the big toe. *4* An ultrasound probe with a 6 cm recording surface was placed on the skin over the belly of tibialis anterior. The probe was oriented to the mid-sagittal plane of tibialis anterior and used to image the muscle during the contractions. *5* The average EMG of the tibialis anterior, medial and lateral gastrocnemius, and soleus muscles was recorded with surface electrodes attached to the surface of the skin over muscles. **b** An example of an ultrasound image of the tibialis anterior from one young subject (*left*) and a corresponding schematic of the features identified from the image to determine the architectural features of the muscle: *a* superficial aponeurosis of the tibialis anterior; *b* the muscle fascicle extending from the central to the superficial aponeurosis; *c* the angle between the fascicle and the superficial aponeurosis; and *d* the angle between the fascicle and central aponeurosis

Ultrasound images obtained during the contractions were synchronized with the EMG and kinematic measures by an external, custom-made trigger (University of Iowa) that was manually activated just prior to the start of each contraction.

Experimental protocol

Subjects began the experiment by performing maximal voluntary contractions (MVC) with the dorsiflexors and plantar flexors to determine the maximal EMG activity of the lower leg muscles. Subjects were provided with a restraint on the dorsum of the foot for the dorsiflexion contractions and with restraints on the plantar aspect of the foot and on the distal end of the upper leg for the plantar flexion contractions. Subjects performed two sets of the maximal isometric dorsiflexion and plantar flexion contractions with approximately 60 s of rest between contractions. Subjects were instructed to achieve a maximal

contraction in 3 s and to hold the contraction for another 3 s. Strong verbal encouragement was provided during the maximal contractions. The maximal value achieved during each task was recorded.

Then 1-RM contractions were performed to determine the maximal load subjects could lift once with the dorsiflexor muscles. The range of motion extended from full plantar flexion to 90° of ankle flexion. Subjects were instructed to lift the load in approximately 1 s and to isolate the dorsiflexor muscles as much as possible when lifting the load. As the dorsiflexors can produce greater force when the ankle is plantar-flexed (Maganaris 2001), beginning the movement from full plantar flexion did not limit the load that subjects could lift with the foot. The initial load varied for subjects, but was usually between 2 and 4 kg. The initial load was increased based on verbal feedback from subjects about how much effort was required to lift the load. The load was increased until the subject either could not lift the load or extended the knee to assist with the task. The load was then decreased in increments of 0.2–0.5 kg until the maximal load that the subject could lift with the foot was determined. Subjects were provided with approximately 60 s of rest between the contractions. Approximately six trials (range: 4–10) were required to identify the 1-RM load.

The experimental task required subjects to lift (shorten the dorsiflexor muscles) and lower (lengthen the dorsiflexor muscles) a load of 15% of 1-RM with the foot over a 15° range of motion. The 15% 1-RM load was used to ensure that all subjects could achieve the fastest movement speed and because differences in motor-output variability between young and old adults are most obvious at low muscle forces (Barry et al. 2007; Burnett et al. 2000; Graves et al. 2000; Laidlaw et al. 2000, 2002). As subjects began the task by supporting the load with the ankle extended at 105°, each trial began with an isometric contraction. Subjects were required to lift and lower the load at speeds of 3, 15, 30, and 50°/s, which corresponded to movement durations of 5, 1, 0.5, 0.33 s, respectively. The speed of 50°/s was not maximal for any subject, but approached the top speed that could be managed by some of the older adults. Additional support, when necessary, was provided to the lower leg by one of the investigators to prevent motion about the knee joint. The angle that the foot made with the lower leg was displayed as a line on a 17-in monitor placed in front of the subjects. A stationary horizontal black line in the middle of the screen represented the ankle angle of 105°. The screen view extended ±15° in the vertical direction and 10° in the horizontal direction. The line representing the ankle angle began on the left side of the computer screen at the start of each trial.

To become familiar with the experimental task, each subject performed several practice trials (total: ~12) at three randomly chosen target speeds. After each trial, subjects were provided with visual and verbal feedback of the estimated movement speed. Subjects were required to repeat the contractions until the estimated speed was within ±15% of the target speed for three trials, but were limited to 15 attempts to match the target speed. Subjects were provided with at least 30 s of rest between trials and at least 60 s between target speeds. The order of the contraction speeds was assigned randomly and the contraction type (shortening or lengthening) was counterbalanced across age groups.

Data analysis

The trials that were estimated to match each target speed were reanalyzed offline and a tolerance of ±20% was used to discriminate between successful and unsuccessful trials. A value of 15 for the number of trials required to complete the task was assigned to subjects who could not match the target speed 3 times in 15 trials.

The acceleration of the foot and the average amplitude of the rectified EMG (aEMG) were quantified over the middle two-thirds of the successful trials. The SD of acceleration of the foot was averaged across the successful trials and normalized to the mass of the load lifted. The stationarity of the acceleration signal was examined over the middle two-thirds of the movement for a subset of trials. The analysis indicated that the mean and variance of the acceleration did not change with time; therefore, the acceleration signals were not de-trended prior to analysis. The aEMGs of the muscles over the middle two-thirds of the successful trials were expressed as a percentage of the MVC value. The EMG value for each muscle from the MVC trials was defined as the maximal aEMG value within a 1-s sliding window that was advanced in one-sample steps. The aEMGs from the soleus and medial and lateral gastrocnemius were averaged to represent the aEMG of the antagonist muscles.

The ultrasound images were quantified using the image analysis software Image J, version 1.41o (National Institutes of Health, Bethesda, MD, USA). Muscle fascicles were identified as the dark spaces between the echoes of the connective tissue. The change in fascicle length was calculated as the difference in fascicle length between the values obtained at 12.5° and 2.5° of foot displacement. The same procedure was used to determine the change in pennation angle during the contraction.

The pennation angle of the muscle fascicle was defined as the average angle the fascicle made between the echoes of the central and superficial aponeuroses. In <4% of instances, the fascicles or aponeuroses were curved. When the fascicles were curved, the area between the aponeuroses was divided into three equal sections parallel to the

aponeuroses. The straight line connecting the aponeuroses and the intersection of the fascicle and the nearest line, which divided the muscle into the three separate areas, was used to calculate the fascicle pennation angle (Maganaris et al. 1998). A similar method was used to calculate pennation angle when an aponeurosis was curved. It was assumed that the muscle fascicles extended linearly from the central to the superficial aponeurosis. As a portion of the fascicle typically extended off the screen, the remaining length was calculated using trigonometry (Fig. 1b). The changes in muscle fascicle length and pennation angle for each trial from a subject were plotted against each other with a linear regression line and the corresponding 95% confidence interval bounds. Values that were outside the 95% confidence interval were excluded from the data set as they likely represented measurement error.

Statistical analysis

A three-factor, linear, mixed-model ANOVA (2 age groups × 2 contraction types × 4 target speeds) with repeated measures on contraction type and target speed was used to compare the dependent variables (SD of foot acceleration, tibialis anterior and antagonist muscle aEMG, and the change in muscle fascicle length and pennation angle) between young and old adults. Most dependent variables did not differ between young and old adults; therefore, the data were collapsed across the age groups and a two-factor, linear, mixed-model ANOVA (2 contraction types × 4 target speeds) with repeated measures on contraction type and target speed was used to compare the dependent variables between the contraction types. Post-hoc tests with Bonferroni correction were used to identify differences between the dependent measures across the levels of target speed when there was a significant main effect for speed. Paired samples *t* tests with a modified Bonferroni correction were used to test differences in the dependent measures between contractions types and speeds when there were significant interactions.

Linear bivariate regression analysis was used to determine the association of the SD of foot acceleration for the shortening and lengthening contractions with (1) movement speed, (2) aEMG of the tibialis anterior, (3) aEMG of the antagonist muscles, (4) change in tibialis anterior fascicle length, and (5) change in tibialis anterior fascicle pennation angle. Linear bivariate regression analysis was also used to determine the associations for the change in tibialis anterior fascicle length and pennation angle during the shortening and lengthening contractions with the aEMG of the tibialis anterior and antagonist muscles. The strength of the linear associations is reported as Pearson product-moment correlation coefficients (r^2). Stepwise multiple linear regression analyses were used to determine the contribution of each independent variable (movement speed, aEMG of the tibialis anterior, aEMG of the antagonist muscles, and changes in muscle fascicle length and pennation angle) to the prediction of the SD of foot acceleration for the shortening and lengthening contractions. The goodness-of-fit for the multiple regression model is reported as the squared multiple correlation (R^2). The relative importance of each variable to predicting the SD of foot acceleration when the linear effects of the other variables were accounted for is reported as the partial correlation (r). A positive sign for the partial correlation indicates that the predictor and SD of foot acceleration were positively related, whereas a negative sign indicates that they were inversely related. An alpha level of 0.05 is considered significant and values are presented as mean ± SD in the text and mean ± SE in the figures. Statistics were performed in SPPS (version 17.0; SPSS, Chicago, IL, USA).

Results

Young and old adults performed shortening and lengthening contractions with the dorsiflexor muscles at four target speeds. The 1-RM load that young adults could lift with the foot (7.43 ± 1.73 kg) was significantly greater than that for old adults (5.5 ± 1.51 kg; $P = 0.005$). The SD of foot acceleration increased with movement speed ($P < 0.001$), but the change in fascicle length and pennation angle was not significantly associated with the SD of foot acceleration for the shortening ($r^2 \leq 0.031$; $P \geq 0.092$) or lengthening contractions ($r^2 \leq 0.031$; $P \geq 0.101$). The data were collapsed across age due to the absence of a difference between young and old adults ($P = 0.093$) for the SD of foot acceleration. Representative data for foot acceleration, tibialis anterior aEMG, and change in foot angle for the shortening and lengthening contractions are shown for a young subject in Fig. 2.

Kinematics

The number of trials performed to achieve three trials that matched the target speed increased with the movement speed ($P < 0.001$). Post-hoc tests revealed that more trials were needed for the target speeds of 15, 30, and 50°/s than for 3°/s ($P \leq 0.004$) and that the number of trials needed was greater for the lengthening contractions (9.12 ± 3.81) than for shortening contractions (7.68 ± 3.50; $P = 0.001$). In addition, old adults performed more trials (9.13 ± 4.03) than young adults (7.57 ± 3.19) to match the target speed ($P = 0.001$). There was no interaction between any of the factors (age × contraction type: $P = 0.866$; age × speed: $P = 0.125$; speed × contraction type: $P = 0.354$; Table 1).

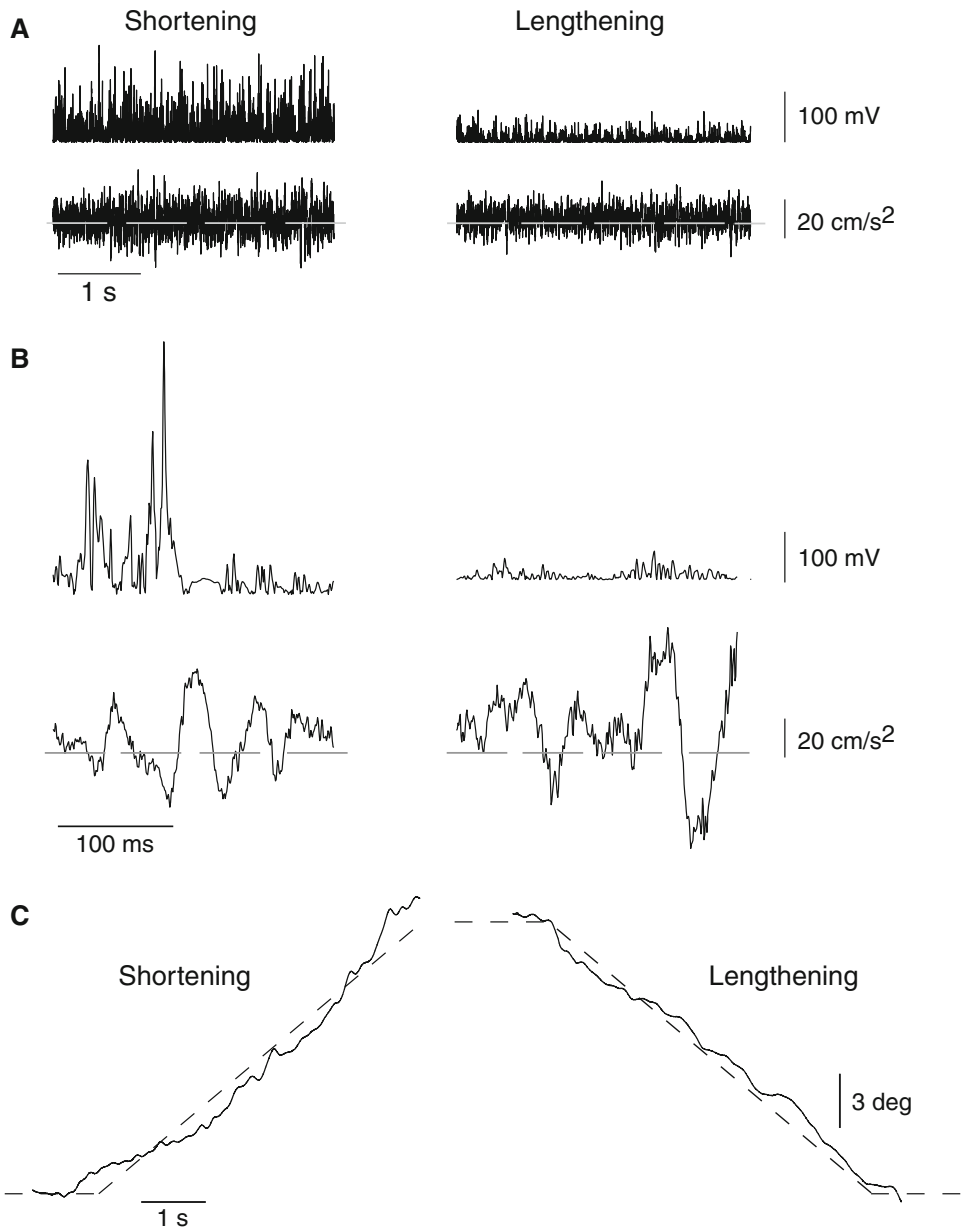

Fig. 2 Example data for the shortening and lengthening contractions performed by a young subject at a target speed of 3°/s (**a**) and 50°/s (**b**). The top trace in **a** and **b** corresponds to the rectified EMG of the tibialis anterior muscle during the middle two-thirds of the contraction and the bottom trace denotes the acceleration of the foot over the same period. *Dashed lines* through the acceleration data indicate zero acceleration. **c** An example of the angle of the foot relative to the lower leg during shortening and lengthening contractions performed by the same young subject at a target speed of 3°/s. The target angle is represented as the *dashed line* and the *solid line* indicates the subject's angle. Each contraction was preceded by an isometric contraction

Table 1 Number of trials performed at each target speed (3, 15, 30, and 50°/s) to achieve three trials that matched the criterion speed

	Shortening				Lengthening			
	3	15	30	50	3	15	30	50
Young	5.39 (1.66)	7.62 (2.66)	7.92 (2.81)	6.54 (3.21)	5.46 (1.71)	7.85 (2.79)	10.4 (3.57)	9.39 (3.53)
Old	5.71 (3.12)	8.07 (3.39)	8.71 (4.41)	10.9 (3.32)	7.57 (3.52)	8.71 (4.01)	11.5 (3.70)	11.9 (3.23)

Mean (SE). The number of trials increased with the speed of the contraction ($P < 0.001$). More trials were needed for the target speeds at 15, 30, and 50°/s, than at 3°/s ($P \leq 0.004$). The number of trials to match the target speed was greater for the lengthening contractions than for shortening contractions ($P = 0.001$). Also, old adults performed more trials than young adults to produce three trials that matched the target speed ($P = 0.001$)

The SD of foot acceleration increased with the speed of the movement ($P < 0.001$). Post-hoc tests revealed that the SD of foot acceleration differed between each contraction speed ($P \leq 0.016$, Bonferroni correction). However, the SD of foot acceleration did not differ between the shortening (12.8 ± 7.88 cm s^{-2}/kg) and lengthening contractions

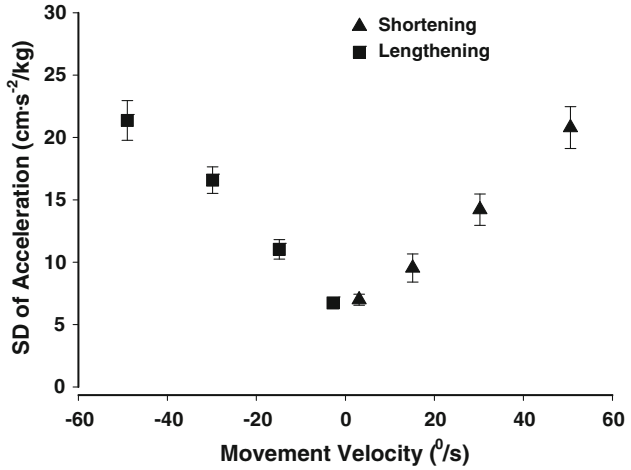

Fig. 3 The SD of foot acceleration for the middle two-thirds of each trial plotted as a function of movement velocity for the shortening (*triangle*) and lengthening (*square*) contractions. A negative movement velocity denotes lengthening contractions and a positive velocity indicates shortening contractions. The SD of acceleration did not differ between the shortening and lengthening contractions ($P = 0.392$). Values are mean ± SE

(13.7 ± 7.49 cm s^{-2}/kg; $P = 0.392$) and increased with movement speed similarly for shortening and lengthening contractions ($P = 0.502$; Fig. 3).

Agonist and antagonist EMG activity

The aEMG for tibialis anterior increased as target speed increased ($P < 0.001$) and was greater for shortening contractions (27.2 ± 11.7% MVC) than for lengthening contractions (9.73 ± 5.21% MVC; $P < 0.001$; Fig. 4a). The aEMG for tibialis anterior increased from 19.5 ± 6.54 to 37.9 ± 12.9% MVC with movement speed for the shortening contractions, but decreased from 12.2 ± 5.41 to 7.78 ± 4.63% MVC with movement speed for the lengthening contractions (contraction type × speed interaction: $P < 0.001$). Paired samples *t* test indicated that the aEMG for tibialis anterior was greater for the shortening contractions than for the lengthening contractions at each movement speed ($P < 0.001$, modified Bonferroni).

The aEMG of the antagonist muscles (average of medial and lateral gastrocnemius and soleus) increased with target speed ($P = 0.010$; Fig. 4b) similarly for the shortening (range: 1.99–59.7% MVC) and lengthening contractions (range: 1.59–68.7% MVC; speed × contraction type interaction: $P = 0.088$). The antagonist aEMG did not differ between the shortening (7.77 ± 7.63% MVC) and lengthening contractions (9.56 ± 10.3% MVC; $P = 0.304$).

Change in muscle architectural parameters

The amount of change in fascicle length did not differ across target speeds ($P = 0.385$; Fig. 5a). The decrease in fascicle length during shortening contractions (0.937 ± 0.633 cm) was greater than the increase in fascicle length during lengthening contractions (0.806 ± 0.521 cm; $P = 0.014$), but did not differ across movement speeds (contraction type × speed interaction: $P = 0.763$).

The change in fascicle pennation angle did not differ across movement speeds ($P = 0.516$; Fig. 5b). The increase in pennation angle of tibialis anterior fascicles during shortening contractions (1.61 ± 0.918°) was greater than the decrease in pennation angle during lengthening contractions (0.966 ± 0.632°; $P < 0.001$), but did not differ across movement speeds (contraction type × speed interaction: $P = 0.811$).

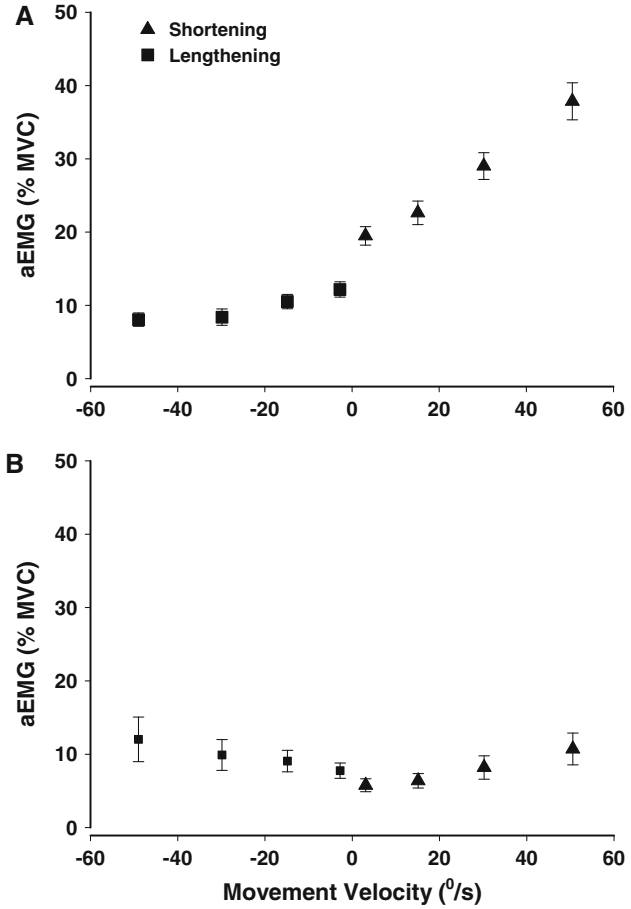

Fig. 4 The aEMG for the middle two-thirds of each trial plotted as a function of movement velocity for the shortening (*triangle*) and lengthening (*square*) contractions. A negative movement velocity denotes lengthening contractions and a positive velocity indicates shortening contractions. **a** The aEMG of tibialis anterior was greater for the shortening contractions than for lengthening contractions at each movement speed ($P < 0.001$). **b** The aEMG of the antagonist muscles (average for soleus and medial and lateral gastrocnemius) increased with movement speed ($P = 0.01$). The aEMG of the antagonist muscles did not differ between the shortening and lengthening contractions ($P = 0.304$). Values are mean ± SE

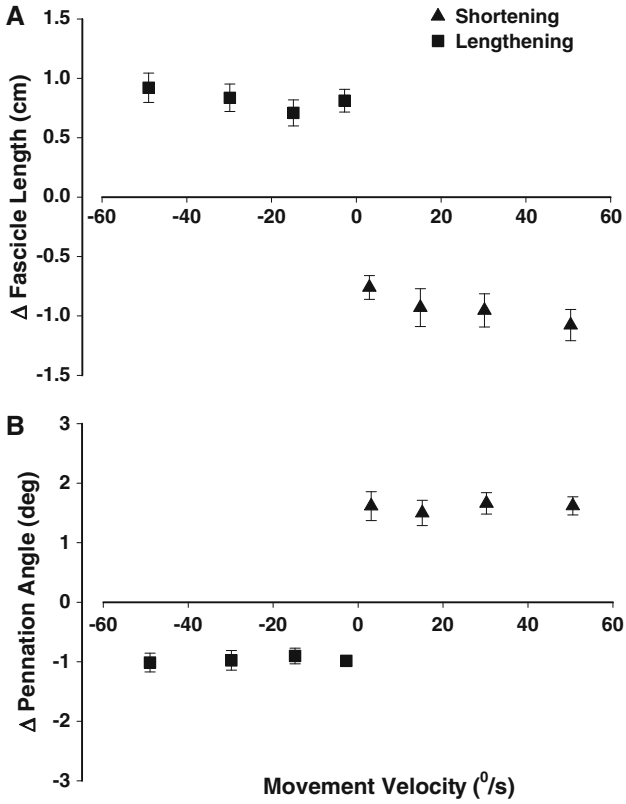

Fig. 5 The change in muscle architectural parameters over the middle two-thirds of the contractions plotted as a function of movement velocity for the shortening (*triangle*) and lengthening (*square*) contractions. A negative fascicle length value denotes fascicle shortening and a negative pennation angle indicates a decrease in angle. **a** The increase in length of tibialis anterior fascicles during the lengthening contractions was on average less than the decrease in length during the shortening contractions ($P = 0.014$). **b** The decrease in pennation angle of tibialis anterior fascicles during the lengthening contractions was on average less than the increase in pennation angle during shortening contractions ($P < 0.001$). Values are mean ± SE

The change in tibialis anterior fascicle length was associated with aEMG of the tibialis anterior, but the association was weak and only evident during the shortening contractions ($r^2 = 0.05$; $P \leq 0.041$). However, the change in tibialis anterior pennation angle was associated with both the aEMG of the tibialis anterior and antagonist muscles during the shortening contractions ($r^2 \geq 0.05$; $P \leq 0.031$). The changes in tibialis anterior fascicle length and pennation angle were not associated with the aEMG of the tibialis anterior or antagonist muscles during the lengthening contractions ($r^2 \leq 0.03$; $P \geq 0.131$).

Associations with the SD of foot acceleration

The SD of foot acceleration increased as the speed of foot displacement increased for the shortening ($r^2 = 0.414$; $P < 0.001$) and lengthening contractions ($r^2 = 0.554$; $P < 0.001$). The aEMG of tibialis anterior was positively ($r^2 = 0.123$; $P < 0.001$) and inversely ($r^2 = 0.105$; $P = 0.001$) associated with the SD of foot acceleration for the shortening and lengthening contractions, respectively. The aEMG of the antagonist muscles was positively associated with the SD of foot acceleration for the shortening contractions ($r^2 = 0.127$; $P < 0.001$), but not the lengthening contractions ($r^2 = 0.008$; $P = 0.370$). The change in fascicle length and pennation angle was not associated with the SD of foot acceleration for either shortening (fascicle length: $r^2 = 0.031$; $P = 0.092$; pennation angle: $r^2 = 0.003$; $P = 0.605$) or lengthening contractions (fascicle length: $r^2 = 0.018$; $P = 0.215$; pennation angle: $r^2 = 0.031$; $P \geq 0.101$).

Stepwise multiple linear regression analysis was used to examine the change in the SD of foot acceleration across the shortening and lengthening contractions. The purpose of the analysis was to determine if the inclusion of tibialis anterior aEMG and antagonist aEMG as predictor variables would increase the strength of the association between movement speed and the SD of foot acceleration. The changes in muscle fascicle length and pennation angle were not included in the analysis, as they were not associated with the changes in the SD of foot acceleration (Fig. 3 vs. Fig. 5). Similarly, the antagonist aEMG was not included in the stepwise multiple linear regression for the lengthening contractions. The prediction of the SD of foot acceleration for the shortening contractions improved significantly from its association with movement speed with a multiple linear regression function ($R^2 = 0.453$; $P < 0.001$) that added antagonist aEMG (partial correlation: $r = 0.258$; $P = 0.008$). In contrast, the prediction of the SD of foot acceleration for the lengthening contractions was not improved significantly from its association with movement speed by including the tibialis anterior aEMG as a predictor variable (partial correlation: $r = -0.037$; $P = 0.711$).

Discussion

The motor-output variability of the participants attempting to match a target speed while lifting and lowering a light load with the foot increased with movement speed, but to a similar extent for both young and old adults. The increase in motor-output variability, which was quantified as the SD of foot acceleration, was not associated with the change in fascicle length and pennation angle during either the shortening or lengthening contractions. The agonist aEMG did not improve the prediction of the SD of foot acceleration for either the shortening or lengthening contractions. In contrast, the antagonist aEMG (medial and lateral gastrocnemius and soleus) improved the prediction of the SD

of foot acceleration for the shortening contractions, but not for the lengthening contractions.

Kinematics

The number of trials needed to produce three successful trials increased with target speed, and was greater for the lengthening contractions and for old adults. Christou et al. (2003) also found that a greater number of trials was required for lengthening contractions and for old adults, but there was no effect of target speed, when young and old adults lifted and lowered a submaximal load with the index finger at target speeds ranging from 1.72 to 66.5°/s.

Similar to the results of Christou et al. (2003), the SD of foot acceleration did not differ between young and old adults. These results, however, differ from those reported by others who found old adults had greater fluctuations in index finger acceleration (Burnett et al. 2000; Laidlaw et al. 2000; Laidlaw et al. 2002; Semmler et al. 2000; Semmler 2002) and wrist acceleration (Graves et al. 2000) than young adults during slow shortening and lengthening contractions. The discrepancies could be due to differences in the task requirements. For example, Burnett et al. (2000) required subjects to lift and lower a load with the index finger while matching the angular position of the finger to a triangular template shown on a monitor. In both the current and the Christou et al. (2003) study, subjects performed the contractions without concurrent visual feedback of the limb and target position. Visual feedback has been shown to increase force fluctuations during constant-force contractions, especially for old adults (Tracy 2007), and likely contributed to the age-related impairment in steadiness observed in previous studies (Burnett et al. 2000; Graves et al. 2000; Laidlaw et al. 2000, 2002, Semmler et al. 2000; Semmler 2002). A practice effect might have also contributed to the lack of an age-related difference in the SD of foot acceleration as subjects practiced the tasks until they could achieve the target speed; old adults practiced more than young adults did in the current study and in the one by Christou et al. (2003). Increased practice has been shown to reduce kinematic variability and likely contributed to similar SDs of foot acceleration for young and old adults (Christou et al. 2007; Hortobagyi et al. 2001; Keen et al. 1994; Kornatz et al. 2005; Tracy and Enoka 2006).

The SD of foot acceleration was similar for the shortening and lengthening contractions performed with the dorsiflexors, which contrasts with the results from other studies (Burnett et al. 2000; Christou et al. 2003; Christou and Carlton 2002a, b; Graves et al. 2000; Kornatz et al. 2005; Laidlaw et al. 2000; Semmler et al. 2000; Semmler 2002). The greater SD of acceleration for lengthening contractions performed with the first dorsal interosseous muscle has been attributed to differences in motor unit synchronization, coherence in the 2–12 Hz frequency band (Semmler 2002), and motor unit discharge variability (Laidlaw et al. 2000; Pasquet et al. 2006). Although lower discharge rates are associated with greater discharge variability in the first dorsal interosseous (Barry et al. 2007; Moritz et al. 2005), and the discharge rates of motor units in tibialis anterior are reduced during lengthening contractions (Pasquet et al. 2006), it is unknown if discharge variability is greater for motor units in tibialis anterior during lengthening contractions (Jesunathadas et al. 2008). Furthermore, motor neurons that innervate tibialis anterior may not receive the types of synaptic input required to correlate discharge times and thereby contribute to differences during shortening and lengthening contractions. The absence of such differences may contribute to the similar SD of acceleration of the foot during shortening and lengthening contractions in the current study.

Agonist and antagonist EMG activity

Tibialis anterior aEMG increased with movement speed for the shortening contractions, but decreased with movement speed for the lengthening contractions. These finding are similar to those by Christou et al. (2003) and indicate that the faster lengthening contractions required less tibialis anterior motor unit activity. As force fluctuations increase with an increase in the number of active motor units (Hamilton et al. 2004; Moritz et al. 2005), the data seem to suggest that the increase in the SD of foot acceleration with movement speed for the shortening contractions was attributable to the increase in tibialis aEMG. However, such an interpretation is not appropriate as the increase in the SD of foot acceleration was accompanied by a decrease in tibialis anterior aEMG for the lengthening contractions. Furthermore, the inclusion of tibialis anterior aEMG in the multiple regression analysis did not improve the prediction of the SD of foot acceleration for either the shortening or lengthening contractions.

The antagonist aEMG increased with movement speed similarly for the shortening and lengthening contractions. These findings are similar to those of others (Burnett et al. 2000; Graves et al. 2000), but contrast with others who found that antagonist muscle activity was greater during lengthening contractions (Christou and Enoka 2010; Laidlaw et al. 2002; Pasquet et al. 2006). These discrepancies do not seem to depend on the muscle group, task requirements, or task performance (Burnett et al. 2000; Graves et al. 2000; Laidlaw et al. 2002; Pasquet et al. 2006), but likely indicate variability in the amount of antagonist muscle activity across subjects during lengthening contractions (Burnett et al. 2000).

Changes in muscle architecture

The change in length and pennation angle of the muscle fascicles in tibialis anterior during the shortening and lengthening contractions did not differ between young and old adults. This finding differs from the hypothesis that old adults would exhibit greater changes in muscle fascicle length and pennation angle because of presumably sarcopenia-related reductions in the force-generating and rate-of-shortening capacity of muscles in old adults (Narici et al. 2003; Thom et al. 2007). The absence of an age effect for changes in fascicle length and pennation angle might be due to interactions between the more compliant tendons in the old adults than in those of the young adults (Nakagawa et al. 1996; Narici and Maganaris 2006) and a greater reduction in the functional capabilities of muscles in old adults as compared with those of young adults. More compliant tendons have been associated with less shortening of vastus lateralis muscle fascicles during isokinetic contractions with the knee extensors (Fukunaga et al. 1997) and might have reduced the amount of fascicle shortening for old adults in the current study. Thus, the effects of sarcopenia in the muscles of the old adults in the current study did not seem to influence the relative change in muscle fascicle length or pennation angle.

The change in muscle fascicle length and pennation angle did not differ with movement speed, but the change was greater for the shortening than for the lengthening contraction. Thus, similar values for the SD of foot acceleration during the shortening and lengthening contractions were achieved by different changes in the length and pennation angle of muscle fascicles. These findings are not surprising as the muscle exerts less force during lengthening contractions and are similar to those reported by others in which the change in medial gastrocnemius fascicle length during the gait cycle was less during muscle lengthening as compared with muscle shortening (Spanjaard et al. 2009). Less change in fascicle length and pennation angle during lengthening contractions may also be related to greater lengthening of the tendon (less lengthening of muscle fascicles), which has been proposed as a mechanism to prevent injury and to store elastic energy during the stretch–shortening cycles that occur during gait (Cavagna and Kaneko 1977; Fukunaga et al. 2001; Ishikawa et al. 2005; Kawakami et al. 2002).

There was a weak but significant association between the change in fascicle length and tibialis anterior aEMG, as well as between the change in muscle pennation angle and the aEMG of the tibialis anterior and antagonist muscles during the shortening contractions. In contrast, the change in fascicle length and pennation angle was not associated with the aEMG of the tibialis anterior or the antagonist muscles for the lengthening contractions. The weak and insignificant associations are likely due to the large variability in the changes in muscle fascicle length and pennation angle. However, the findings indicate that changes in muscle fascicle length and pennation angle might be better indicators of muscle activity during shortening contractions than during lengthening contractions.

Associations with the SD of foot acceleration

Linear regression analysis revealed strong associations between movement speed and the SD of foot acceleration for young and old adults for both shortening and lengthening contractions. In contrast to the monotonic increases in the SD of foot acceleration with movement speed, tibialis anterior aEMG increased with the SD of foot acceleration during the shortening contractions but decreased during lengthening contractions. Christou et al. (2003) reported a similar association between first dorsal interosseous aEMG and the SD of index finger acceleration when young and old adults lifted and lowered a light load (15% 1-RM) at six target speeds with the index finger. Although the agonist aEMG was associated with movement speed for the shortening and lengthening contractions, it did not improve the strength of the association between movement speed and the SD of foot acceleration in the current study. Conversely, Christou and Enoka (2010) found that the accuracy with which young and old adults could lift and lower a light load (10% 1-RM) to a target location with the index finger ($\sim$25°/s) was influenced by agonist aEMG. However, the SD of index finger acceleration during the movement was not one of the significant predictor variables of differences in accuracy during these tasks. Thus, the average amount of agonist motor unit activity seems limited for predicting motor-output variability.

In contrast to the findings for the agonist aEMG in the current study, the aEMG of the antagonist muscles was only associated with the SD of foot acceleration for the shortening but not for the lengthening contractions. However, there was a significant increase in the strength of the association between movement speed and the SD of foot acceleration when the antagonist aEMG was included in the multiple regression analysis for the shortening contractions. Similarly, Christou and Enoka (2010) found that antagonist aEMG was a significant predictor variable of the errors observed during the index finger tasks, but the contribution was realized during both the shortening and lengthening contractions. However, the relative amount of antagonist aEMG, as quantified with a coactivation index during the middle of the movement did not influence the accuracy of either the lifting or the lowering task. Thus, it seems the aEMG of antagonist muscles may be a better predictor of motor-output variability than the aEMG of the agonist muscles, at least for shortening contractions.

The current study did not find an association between the changes in length and pennation angle of tibialis anterior muscle fascicles and the SD of foot acceleration for either contraction type. The absence of an association may be related to the relatively modest changes in muscle fascicle length and pennation angle over the range of speeds used in the current study. Although Spanjaard et al. (2009) found a significant effect of speed on changes in muscle architecture for the gastrocnemius over a narrow range of speeds, speeds similar to those used in the current study did not influence changes in either tibialis anterior fascicle length or pennation during maximal contractions performed with the dorsiflexors on an isokinetic dynamometer (Reeves and Narici 2003). Over a greater range of speeds, however, there would likely have been different changes in muscle fascicle length and pennation angle for the shortening contractions (Reeves and Narici 2003) and possibly an association between changes in muscle architecture and the SD of foot acceleration. Nonetheless, the increase in the SD of foot acceleration with movement speed in the current study was not associated with the changes in either muscle fascicle length or pennation angle.

Conclusions

The SD of foot acceleration increased similarly with contraction speed for the shortening and lengthening contractions performed with the dorsiflexor muscles by young and old adults. Despite the similar changes in the SD of foot acceleration, the amplitude of tibialis anterior EMG activity increased during shortening contractions but decreased during lengthening contractions. Furthermore, the amplitude of EMG activity for the antagonist muscles increased similarly during both shortening and lengthening contractions. Although the length and pennation angle of muscle fascicles in tibialis anterior changed in opposite directions during the two types of contractions, the changes did not vary across contraction speeds. The only dependent variable that influenced the change in the SD of foot acceleration in addition to movement speed was the amplitude of the antagonist EMG during the shortening contractions. Taken together, these findings indicate that the influence of movement speed on movement variability (SD of foot acceleration) was similar for shortening and lengthening contractions with the dorsiflexor muscles and that despite weaker muscles and the need for more practice, the old adults were able to match the performance of the young adults. These findings may have implications for training protocols aimed at mitigating sarcopenia-related performance deficits in old adults.

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